Regime-Conditional Value-at-Risk on the DFL Brent

Modelling the basis risk of a physical crude desk

Julien — IFP School — June 2026

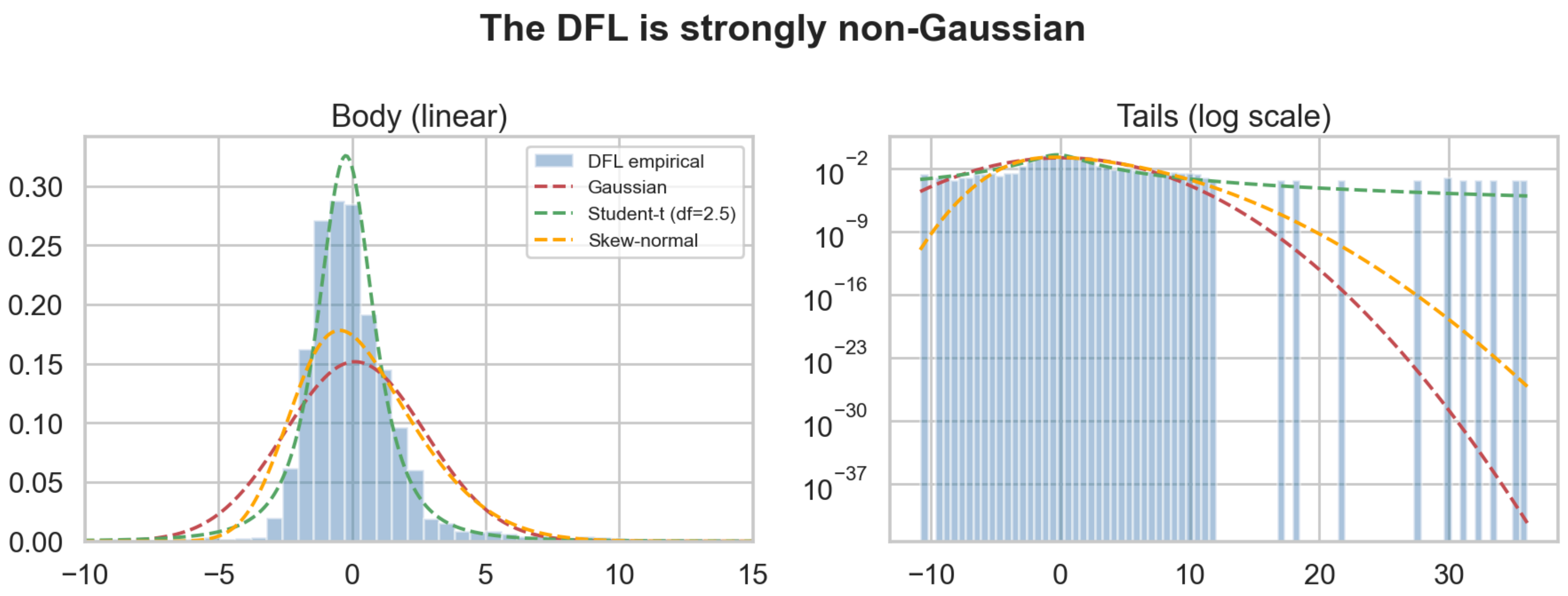
The problem

A physical crude desk buys Dated-priced cargoes and hedges them with ICE Brent futures.

The leftover gap is the DFL — Dated Brent minus front-line ICE Brent. That gap is basis risk: a structural premium that cannot be fully arbitraged away.

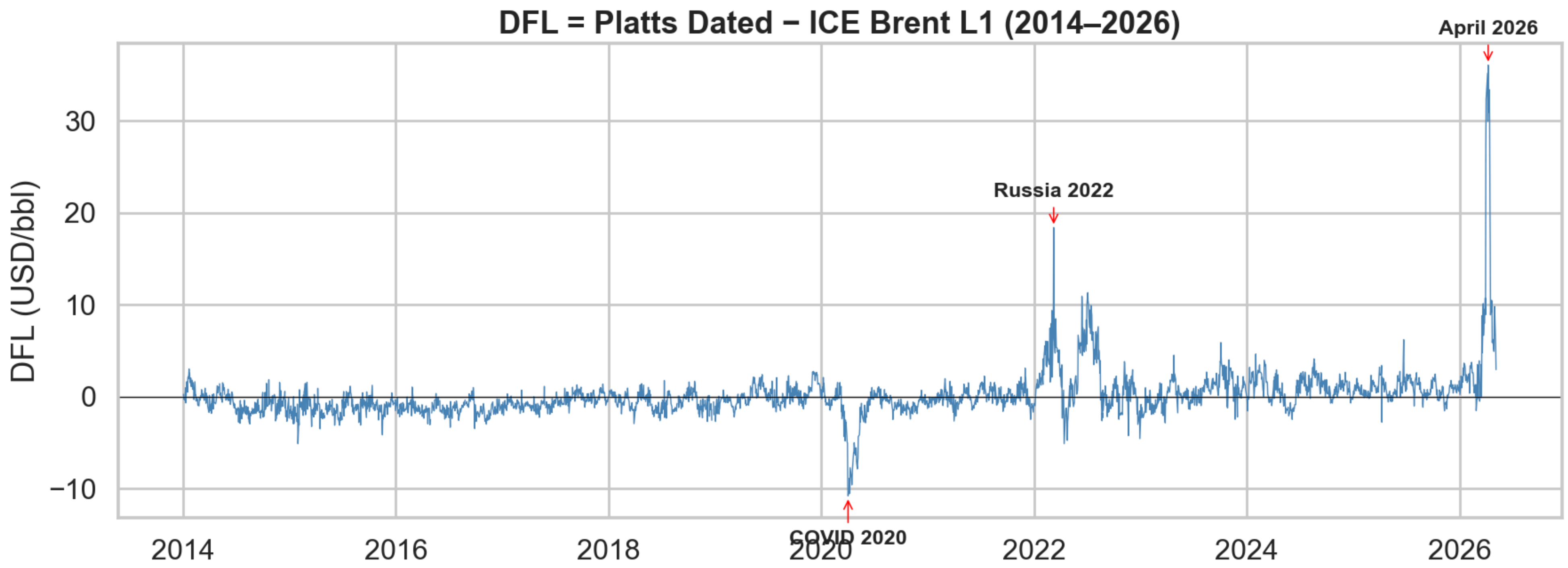
It is real P&L every day, and its tail risk is what we set out to model.

The challenge: the DFL is not normal



Skewness +5.5, kurtosis +62 — extreme, asymmetric fat tails. No single distribution (Gaussian, Student-t, skew-normal) fits, so a flat parametric VaR badly understates the risk.

Why: it is a mixture of regimes



Calm for years, then violent — COVID 2020, Russia 2022, April 2026 (+$36). The extremes cluster, so the DFL behaves differently depending on the market state. The fix: condition the VaR on the regime.

The approach

Data (12 years, 3 crude benchmarks + macro + inventories)

→ PCA — 5 interpretable factors (curve tightness, risk-off, East-West arb, momentum, trans-Atlantic)

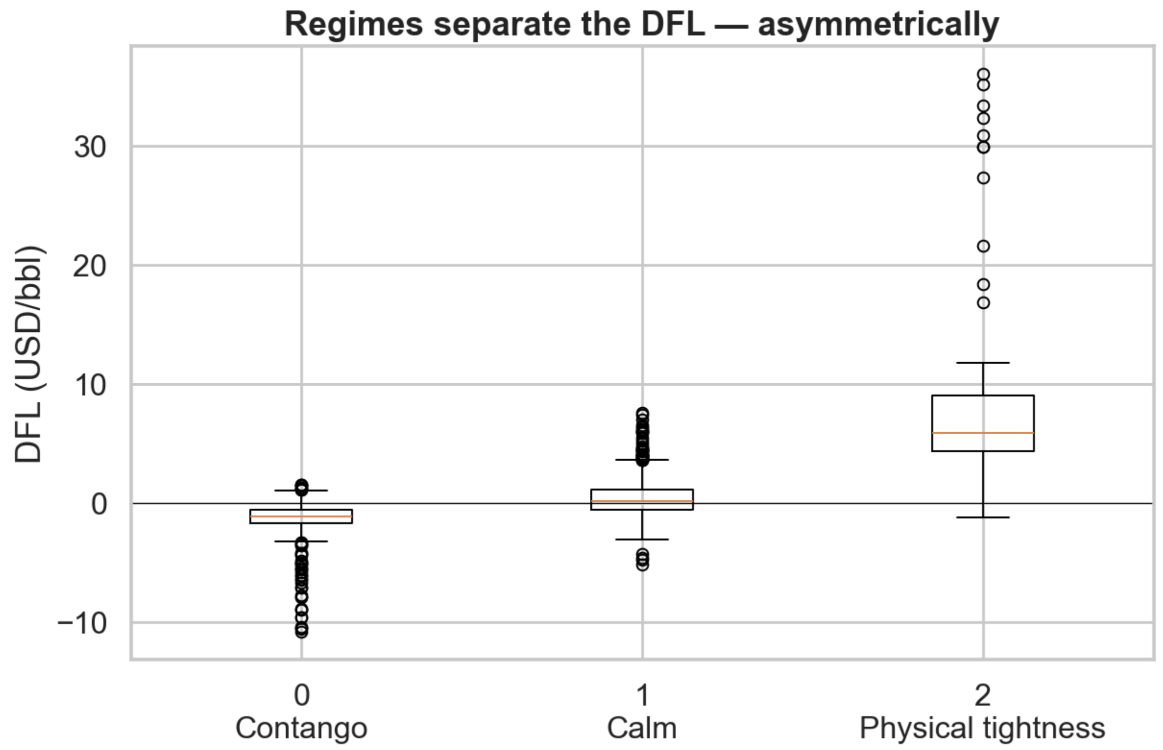
→ K-means — 3 market regimes

→ XGBoost quantile regression — conditional VaR + conformal recalibration

→ Backtest vs 6 baselines, then an ensemble

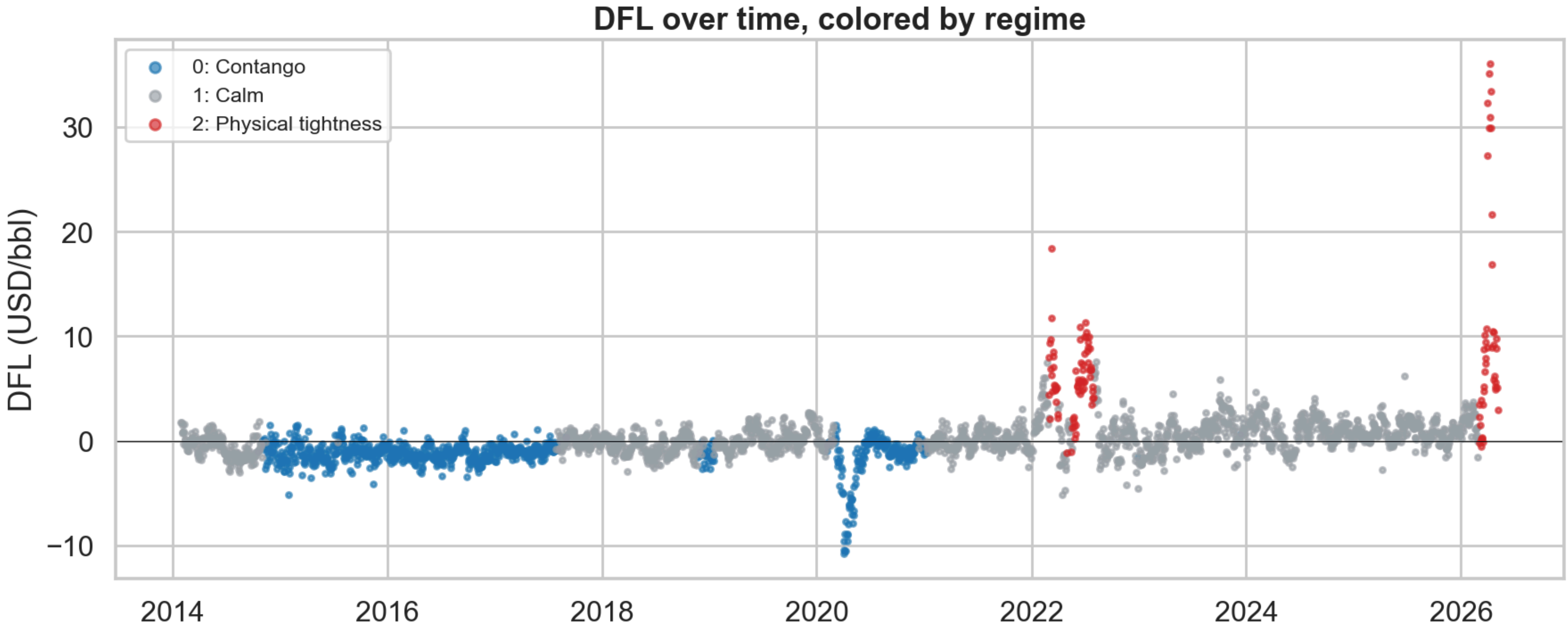
→ Desk application, in dollars

Regimes that mean something



Three directional regimes — Contango (downside), Calm, Physical tightness (upside). Built without ever using the DFL, yet they split it sharply and asymmetrically: contango down to –$11, tightness up to +$36.

The regimes through time



The same regimes laid over the DFL itself: contango (blue) marks the oversupplied, downside periods — including the 2020 COVID dip; tightness (red) the violent upside spikes of 2022 and 2026; calm (grey) everything in between. The unsupervised model rediscovered the market's chronology without ever seeing the DFL.

The model

XGBoost quantile regression on the regime plus continuous market features, predicting the quantiles of the next-day DFL move (q01 / q05 / q95 / q99).

Temporal split: train 2014–2023, test 2024–2026, so the 2026 spike is a genuine out-of-sample stress test.

An upside calibration leak appeared (q95 breached 10.7% vs 5%). After ruling out three causes, it was fixed post-hoc with split-conformal recalibration.

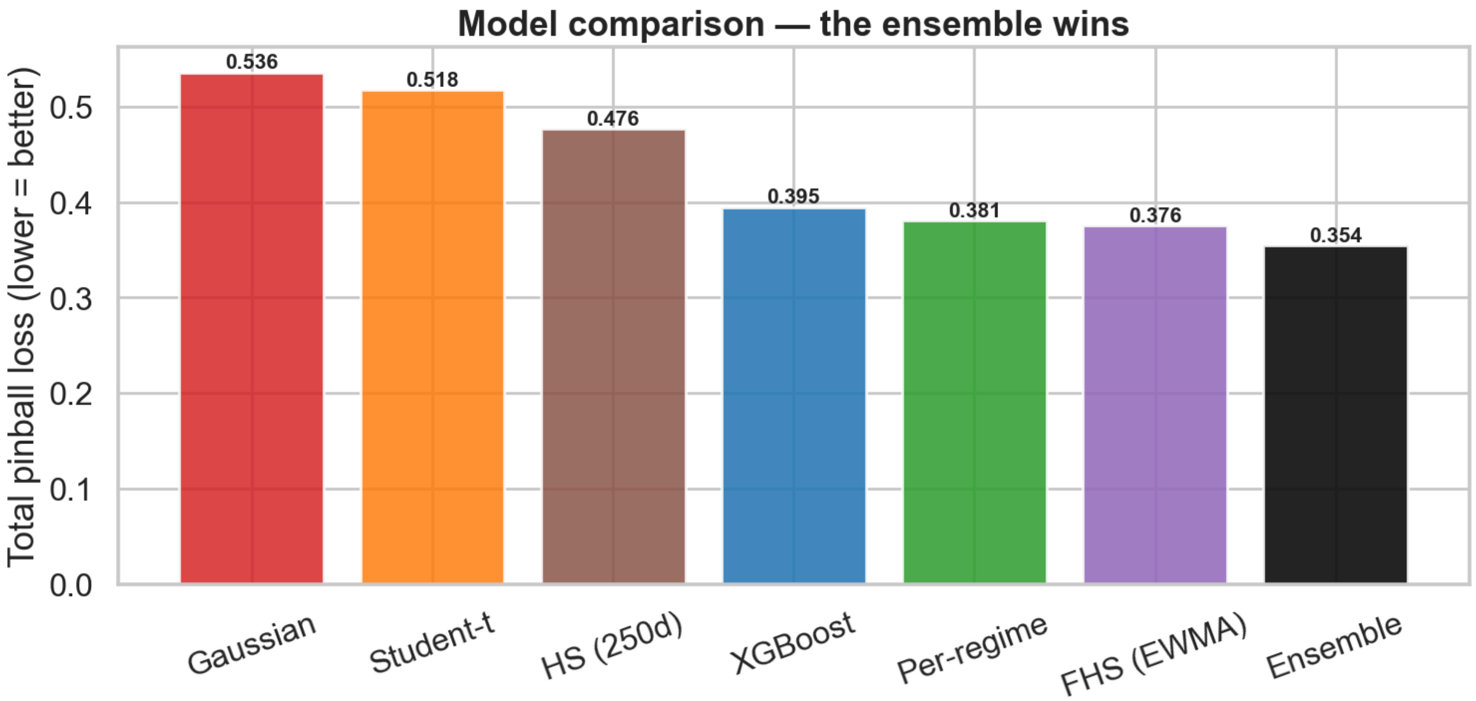
How the models are evaluated

Two axes, not one:

- Calibration — Kupiec (right number of breaches?) and Christoffersen (are breaches clustered?). The regulatory-grade pass/fail.
- Accuracy — pinball loss, the proper scoring rule that ranks the models.

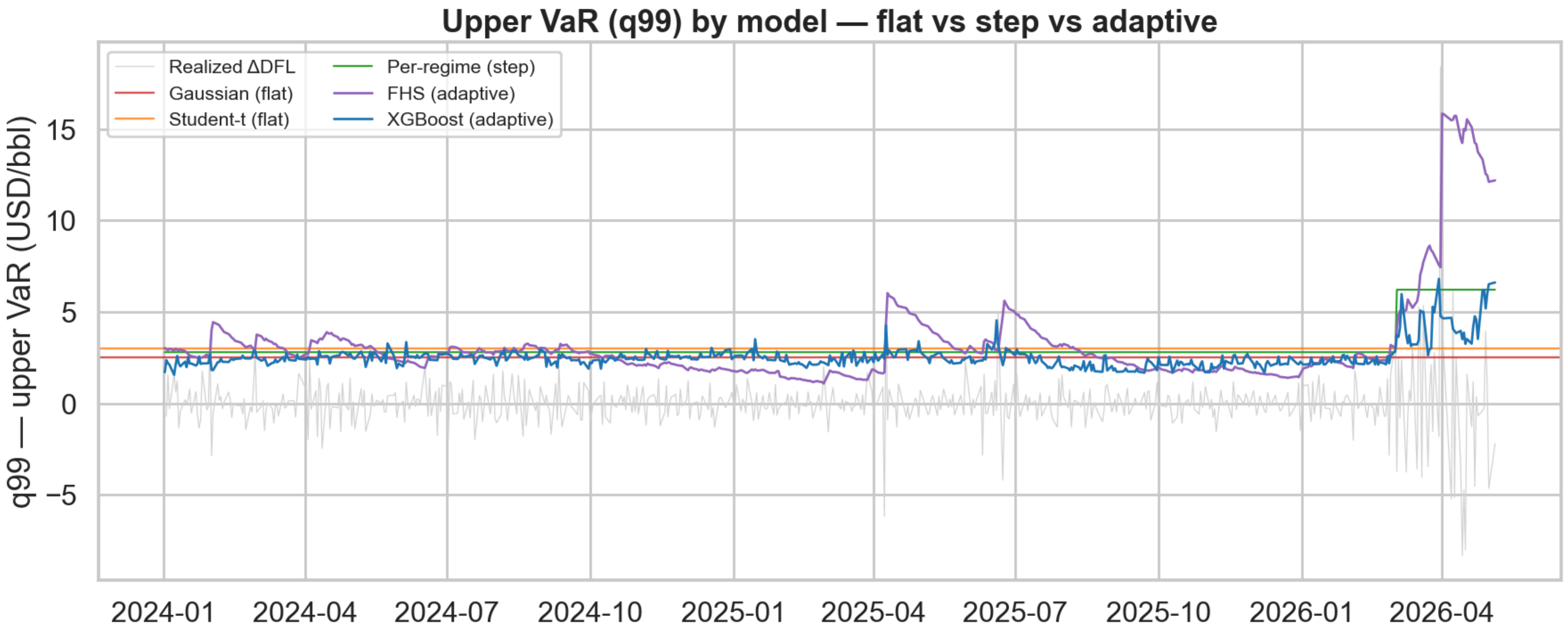
All out-of-sample, against real baselines including Filtered Historical Simulation (the bank standard).

Comparison: who wins



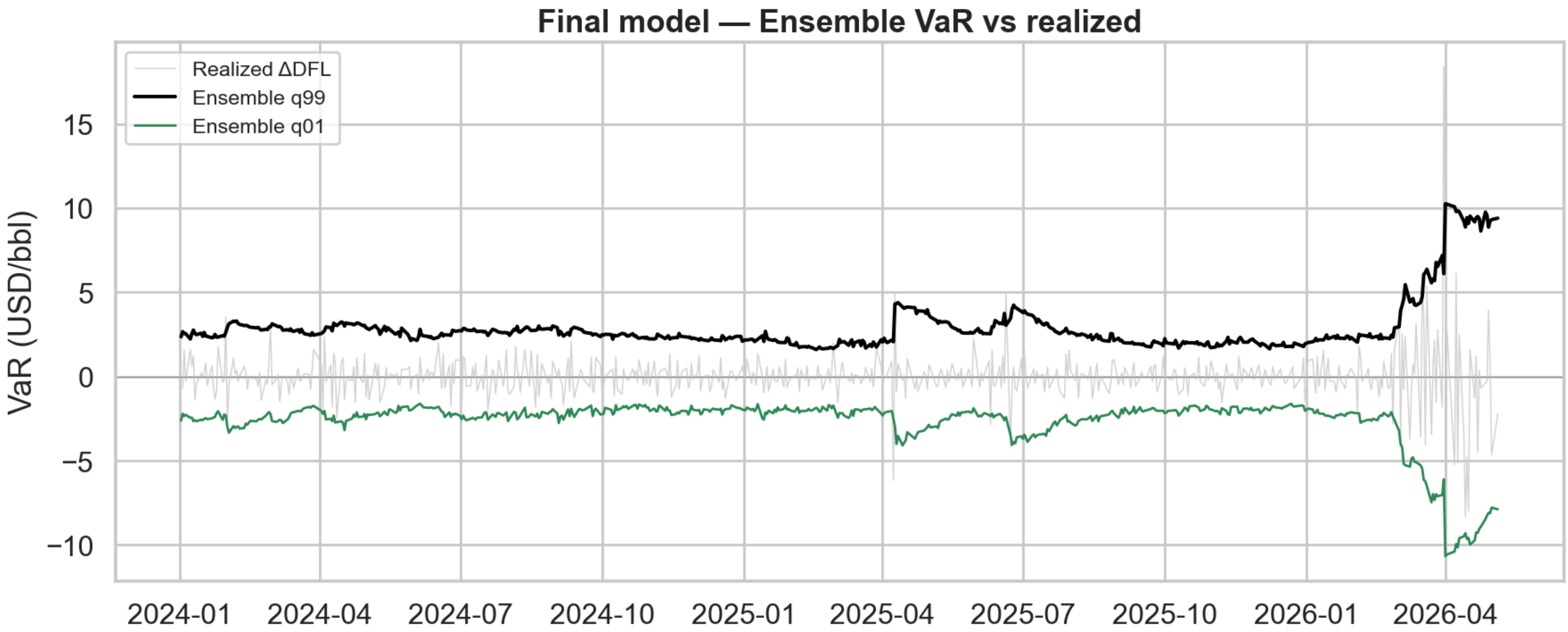
Conditioning beats distribution: Student-t → regime cuts loss by 26%, while adding fat tails buys only 3%. No single model dominates, but the ensemble (XGBoost + FHS) wins and is the only one to pass every backtest.

VaR by model — who adapts



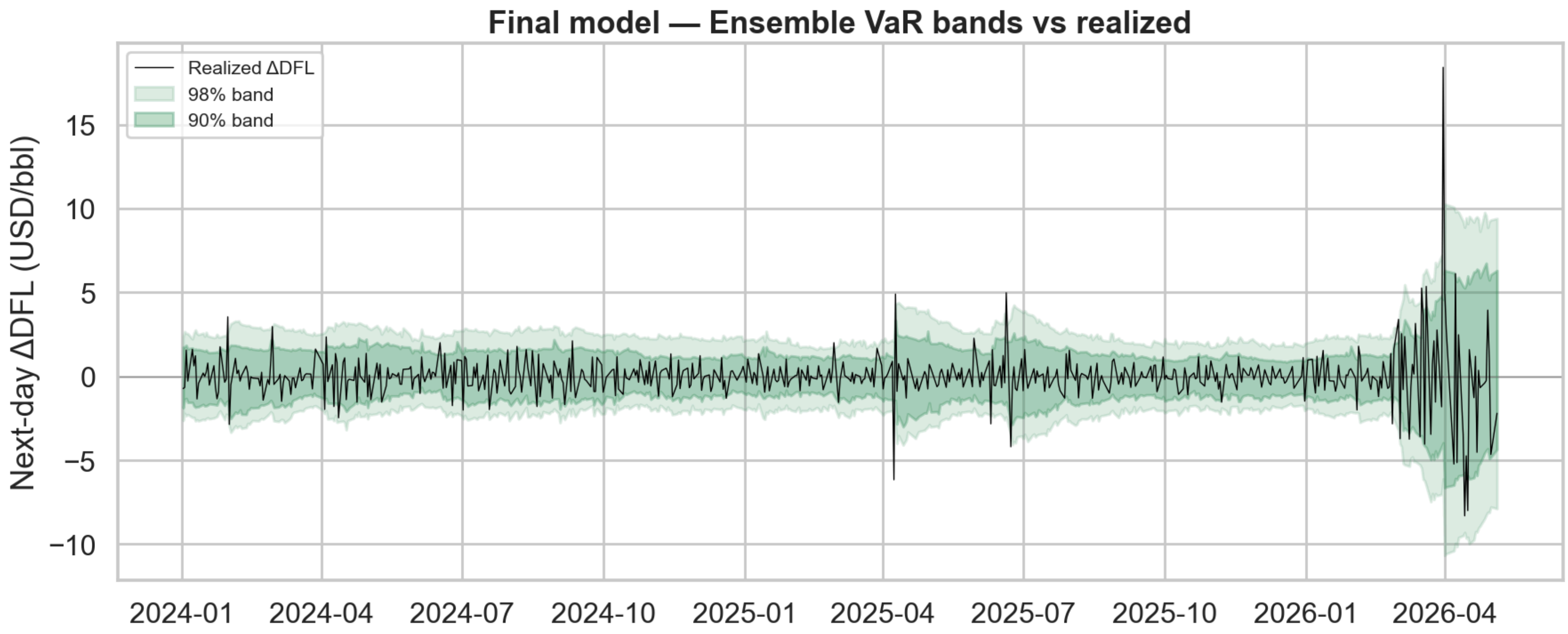
The upper VaR (q99) through time. The parametric models are flat lines, blind to the crisis. The per-regime baseline steps. Only XGBoost and FHS adapt, widening as the April 2026 stress hits. Adaptivity is what separates the top models from the rest.

The ensemble alone



Strip it back to the winning model. The ensemble's VaR sits tight in calm and opens up in the crisis, framing the realized moves on both sides — the same adaptivity as FHS, but tamed, and the only model that passes every backtest.

The final model



Tight in calm, wide in the 2026 crisis — the risk number breathes. Best pinball loss, passes all twelve backtest checks, and fixes the clustered downside breaches that XGBoost alone failed.

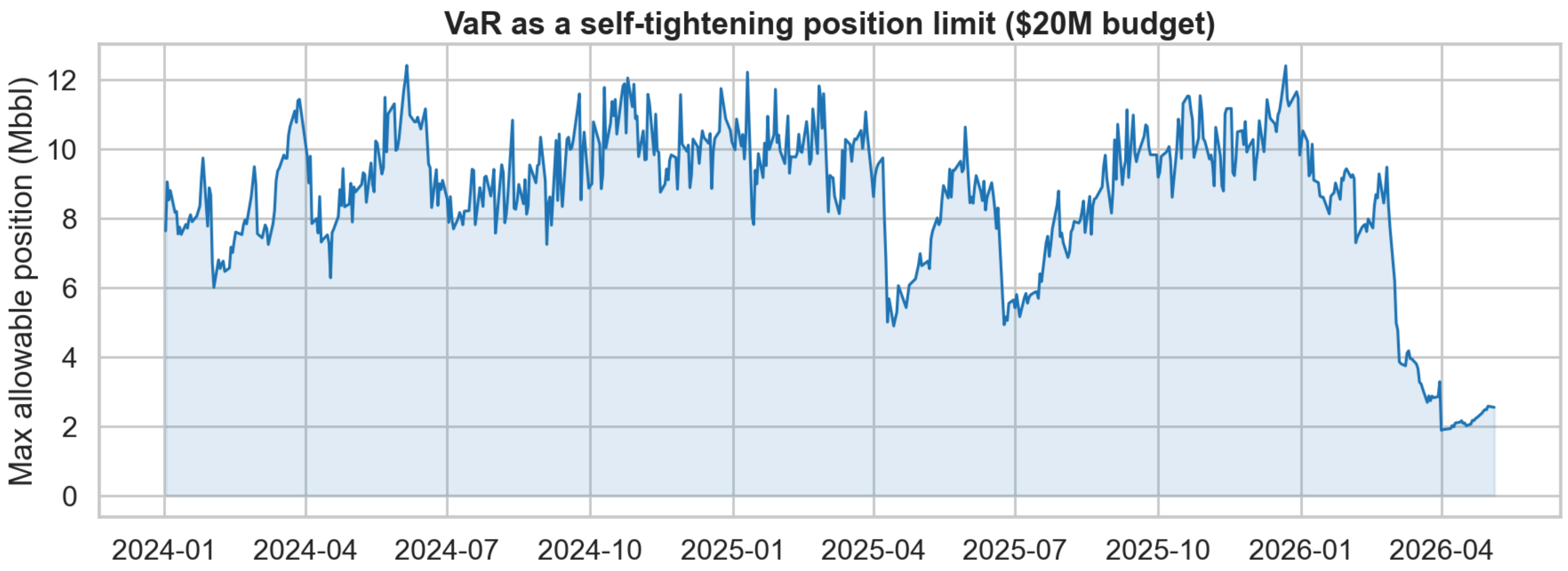
What it means for a desk

For a position long 1,000,000 bbl of DFL:

- 1-day 99% VaR ≈ $2.7M in calm markets, ballooning past $10M in April 2026.
- Expected Shortfall: when the 95% VaR breaks, the average loss is 2.25× the threshold.

The tail is worse than the VaR line suggests — which is exactly why Basel III / FRTB uses Expected Shortfall.

VaR as a self-tightening position limit



Run in reverse: a fixed risk budget sets the allowed position. For a $20M daily VaR budget, the desk can run ~12 Mbbl in calm and under 2 Mbbl in the crisis — a 6–7× cut, automatically, exactly when it matters.

Limits, stated honestly

- The April 2026 move exceeds anything in training — VaR must be complemented by stress testing.
- The regimes are fit on the full sample (look-ahead); production would refit walk-forward.
- Part of the measured daily move is snap-time noise (Platts 16:30 vs ICE 19:30), not real risk.
- The statistically best VaR is not the most usable — FHS's crisis spikes are pro-cyclical.

Takeaways

- An end-to-end regime-conditional VaR, calibrated and formally validated.
- Conditioning on the market state matters far more than the choice of distribution.
- The ensemble is the only fully-calibrated model — structure (XGBoost) plus volatility (FHS).
- Made tangible: dollar VaR, Expected Shortfall, and position sizing.
- And honest about where, and why, it reaches its limits.

THANK YOU

Code and notebooks: github.com/juliens50/ML-project-VaR-Calculation-for-Basis-Risk

Questions?

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